

An Investment Model Based on a Head-And-Shoulder Pattern with Multiple Moving Average Technical Indicators for Future Markets

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ABSTRACT

Future market forecasting is a challenging task in the financial time-series study field. Factors such as the economic environment, political policies, and market news affect it. Although only a few people can profit from it, many researchers and professional investors are eager to spend considerable effort studying the future market in an attempt to develop profitable methods. Certainly, many investment strategies exist in the market including traditional technical analysis and modern emerging artificial intelligence analysis. However, most methods are too complicated or heavily require investors to select and analyze many technical indicators. Moreover, most individual investors do not understand financial statements, the meaning of various technical indicators, and their influence on the future market. These non-professional investors are inevitably in need of a simple and intelligible investment method. Hence, this study proposes a head-and-shoulder combined pattern recognition model based on perceptually important point identification matching (PIP), template matching, and the floating weighted method using two basic technical indicators (5MA and 20MA). A four-year period (2017~2020) of the Crude Oil, NASDAQ100, and S&P500 index was analyzed as the experimental dataset to validate the proposed model. The experimental results show that the proposed model is effective in forecasting accuracy and can be applied to develop investment strategies.

Key words : PIP, template matching, moving average, future market

1. Introduction

Financial market participants generally include institutional and individual investors. It is intractable to profit in the future market especially for common individual investors, and financial market is affected by the economic environment, government policies, and financial interest rates [1-3]. Due to the fact that financial market forecasting is challenging, many investors spend a great deal of endeavor to figure out the method fea-

tured with low-risk, high-return, simple and efficient strategies. According to the Efficient Market Hypothesis [4], asset prices are expected to reflect all available information in the financial market, including historical price information. This implies that historical price data cannot be used to predict future prices. However, with the development of financial and statistical theories, researchers have found various methods in which the financial market does not completely follow the random walk model [5-7]. Meanwhile, the theory of “investor sentiment” in behavioral finance holds that investor viewpoints significantly affect the stock market volatility [8,9]. In addition, many researchers have demonstrated contradictory situations in the financial market using the Efficient Market Hypothesis [10,11]. Therefore, the real-world financial market is ineffi-

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cient and thereby investors can profit from historical price data. Many investment analysis methods have been invented through the tireless study of researchers. For instance, Lo and Mamaysky proposed an approach for technical pattern recognition based on nonparametric kernel regression [12]. Leigh used template matching to implement a variation in bull flag charting to determine trading timing [13]. Lee proposed a knowledge-based data representation method to represent a financial time series containing rich information [14]. Using the fuzzy logic, Naranjo developed a fuzzy trading system for candlestick patterns [15]. Cheng proposed a hybrid forecasting model based on rough sets theory and genetic algorithms using multi-technical indicators [16]. Kim proposed a GA-based method to predict the stock price index [17]. Cao and Tay applied SVM with adaptive parameters in financial time-series forecasting to make a more accurate prediction and solution sparser [18]. Ko used artificial neural networks to determine the best resource allocation in an investment portfolio [19]. Nakano proposed a model to discover trading rules in the Bitcoin trading market based on artificial neural networks [20]. Lee used the Savitzky-Golay filter and LSTM to predict the KOSPI200 index [21]. These methods can be mutually combined to develop investment strategies. However, several drawbacks of these methods should be in consideration: (1) Traditional technical methods use many technical indicators, which mainly rely on the subjective experience of investors to make decisions. Investors need to be able to identify unknown signals in various patterns. However, since people have different sentiments to market charts, they cannot constantly stay in objective and rational standpoint. (2) Many technical analysis methods rely on mathematical models and statistical methods. Most investors find it difficult to learn and use these methods. They do not know which statistical assumptions need to be satisfied and they do not know how to verify these assumptions. (3) The artificial neural network (ANN) builds on a black box inference with many complex parameters and is hard to understand [22]. This paper introduces a combined investment model based on perceptually important point identification matching (PIP), template matching and the floating weighted method using two basic technical indicators (5MA and 20MA). This type of model-based method executes transactions with objective and rational decision rules and these prevent individual investors from making ineffective investment decisions that are often influence by fluctuation in security prices. This model does not use many technical indi-

cators for better understanding, making it possible for common investors to apply. The paper is organized as follows. Section 2 introduces related works. The proposed model is described in Section 3. Section 4 verifies the model performance by providing its results. Section 5 presents the conclusions and introduces the drawbacks of the proposed method.

2. Related Works

This section reviews related works of perceptually important point (PIP) identification matching and template matching.

2.1 Perceptually important point (PIP) identification matching

For a particular chart, the weights of each point are different. On the one hand, not every point has the same importance to the overall shape of the graph, and too many points will cause a rapid increase in the calculation. Otherwise, too many data points will create unnecessary data noise which is obscure and invalid information. Therefore, simplification of the time series and finding is the first problem to address and it is important to represent the time series pattern using these selected critical points. As introduced by Chung [23], PIP achieves this purpose by retaining perceptually important points and is used for the technical analysis of future data, such as time-series pattern matching. For a certain data sequence P , by the identification measure PIP identification matching, the first two PIPs are the starting point P_1 and ending point P_2 of P . The third PIP is the point with the maximum distance from the first two PIPs. The following three methods - Euclidean, per-

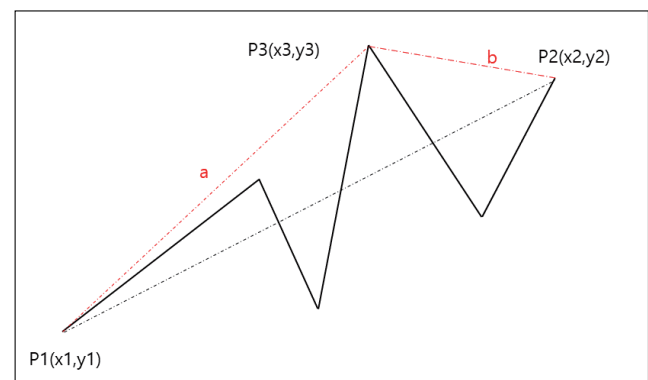


Fig. 1. The Euclidean distance.

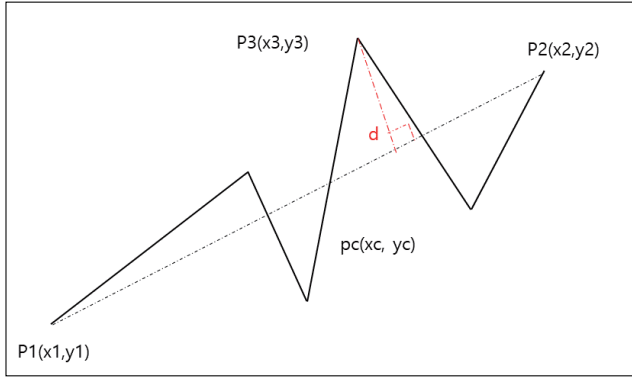


Fig. 2. The perpendicular distance.

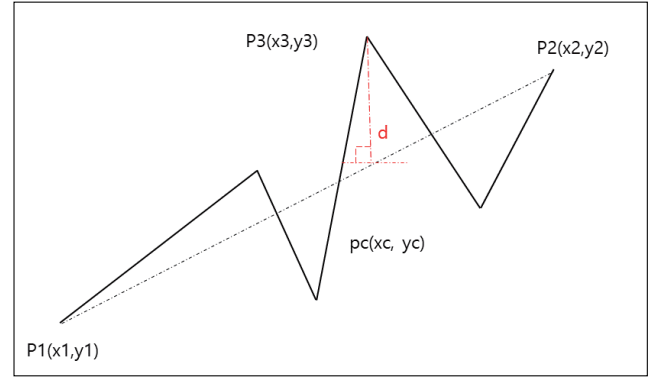


Fig. 3. The vertical distance.

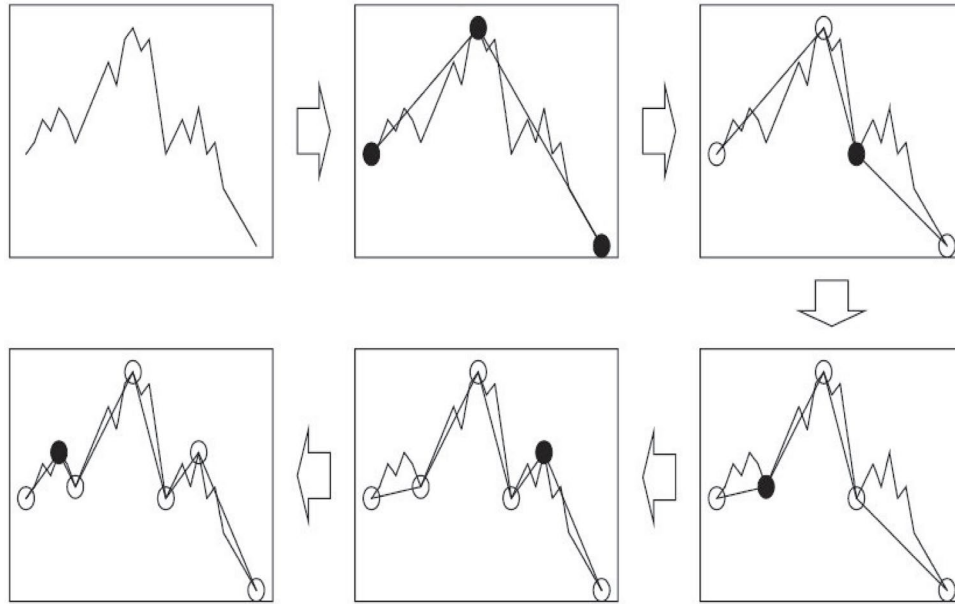


Fig. 4. The identification process of head-and-shoulder pattern.

pendicular, and vertical distances, as shown in Figs. 1~3, can be considered to determine the maximum distance to describe the graphical importance of P_3 relative to P_1 and P_2 . This study used the fastest vertical distance method [24] to determine the perceptually important point.

From Figs. 1 and 2, the distance was determined by calculating the vertical distance from $P_i(x_i, y_i)$ to the line connecting $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$. This type of distance, VD, is defined as follows:

$$\text{Slope} = S = \frac{y_2 - y_1}{x_2 - x_1} \quad (1)$$

$$x_c = \frac{x_3 + (sy_3) + (s^2x_2) - (sy_2)}{1 + s^2} - (x_3)^2 \quad (2)$$

$$y_c = (sx_c) - (sx_2) + y_2 \quad (3)$$

$$VD_{(p_3, p_c)} = |y_c - y_3| = \left| \left(y_1 + (y_2 - y_1) \cdot \frac{x_c - x_1}{x_2 - x_1} \right) - y_3 \right| \quad (4)$$

Based on this method, the process of selecting the most important points is as follows: First, fix the starting point and ending points, calculate the distance of each point to these two points, set the farthest point as a new sampling point and then perform the same steps in the rest of the points, find the point with the largest distance and repeat this step until n points are filled.

The identification process for the head-and-shoulder pattern (Fig. 4) was used to understand the application of the PIP identification method.

2.2 Template matching

Template matching, which is generally used for fitting, is a pattern-recognition technique that can match a given pattern with pictographic images for object recognition. In the financial market, price or volume is usually chosen to predict market trends and make investment decisions [25,26]. In the matching process, the data in the selected period (fitting window) are transformed to a matrix according to the percentile order of data and the matrix and the given template are multiplied to calculate the fit value. The fit value represents the degree of matching between the selected time series data and the given template. The larger the fit value, the higher the matching degree.

Consider bull flag pattern as an example. The flag pattern is a technical chart pattern resembling a flag-shaped pattern, which is a horizontal or sloping flag of consolidation followed by a sharp rise in the positive direction, the breakout [26,27].

The template in Fig. 5 was used to recognize the bull flag pattern. As shown in the feature, the template is a 10×10 grid

0.5	0	-1	-1	-1	-1	-1	-1	-1	0
1	0.5	0	-0.5	-1	-1	-1	-1	-0.5	0
1	1	0.5	0	-0.5	-0.5	-0.5	-0.5	0	0.5
0.5	1	1	0.5	0	-0.5	-0.5	-0.5	0	1
0	0.5	1	1	0.5	0	0	0	0.5	1
0	0	0.5	1	1	0.5	0	0	1	1
-0.5	0	0	0.5	1	1	0.5	0.5	1	1
-0.5	-1	0	0	0.5	1	1	1	1	0
-1	-1	-1	-0.5	0	0.5	1	1	0	-0.2
-1	-1	-1	1	-1	0	0.5	0.5	-2	-2.5
Consolidation					Breakout				

Fig. 5. Bull flag pattern template.

with weights ranging from -2.5 to $+1.0$ in the cells. The weights represent the degree of similarity to a certain pattern. For example, “0.5” means part of the stock pattern falling in the grid has a higher degree of similarity to this pattern than “0”, but lower than “1”. “ -0.5 ” means the part of stock pattern falling in the grid has a lower degree of the similarity than “0”, but higher than “ -1 ” [27,28].

We apply the 10×10 grid to the time series of price or volume data for one trading day at a time, with the leftmost time series data point being the values for the trading day that precedes the current day by 49 trading days if the fitting window is 50 days, and the rightmost time series data point being the trading day that is currently being analyzed. Values for the earliest 10% of the trading days (5 days of the 50 in the rolling window) are mapped to the first column of the grid, values for the next-to-earliest 10% of the trading days are mapped to the second column of the grid and so on, until the most recent 10% of the trading days are mapped to the rightmost column [27,28].

The row in which the trading days fall depends on the percentile order of the price or volume data in the period. The data in the fitting window were sorted from the largest to the smallest. The top 10% of the data (the top six data) correspond to the first row, 11% to 20% correspond to the second row, and so on. The data ranged from 91% to 100%, correspond to the 10th row. For each trading day, we find the grid to which it belongs, and the sum of these weights is the fit value.

3. Proposed Model

This study proposes an investment model based on the head-and-shoulder bottom pattern with two moving average technical indicators. The head-and-shoulder bottom pattern,

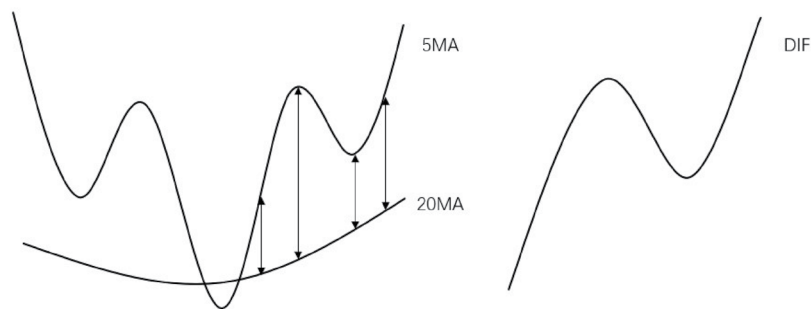


Fig. 6. Head-and-shoulder combined pattern.

sometimes referred to as the inverse head-and-shoulder pattern, is a common and reliable reversal pattern. A head-and-shoulder bottom pattern must have three features, with the middle trough (head) lower than the other two (shoulders). The location points of the left and right shoulders are approximately equal. When this pattern occurs, the price trend changes from a decline to a rise. In addition, to improve the probability of making a profit, as shown in Fig. 6, we add a new pattern to form a combined model using the differential value (DIF) of the technical indicator moving average (MA) on the head-and-shoulder bottom pattern.

Differential Value is defined as

$$DIF = 5MA - 20MA, \quad (5)$$

where 5MA and 20MA represent the short-term and the long-term line respectively.

If after the point of the head, the difference between short-term and long-term lines, DIF maintains an upward trend and can be supported when an adjustment occurs and it can be inferred that in the subsequent market, the future price can continue to rise. Figure 7 indicates the process of the proposed model.

3.1 Template and threshold generating

3.1.1 Head-and-shoulder pattern and 5MA-20MA

combined model pattern recognition

In this process, we used the PIP identification matching method to recognize the 5MA-20MA combined model pat-

terns from historical data. There are two parameter fittings: the Windows and Holding periods that must be set in this method. Fitting-Windows is the length of the transaction term during one test. The holding period represents the duration of the stock held by investors after the last testing day. Table 1 contains the

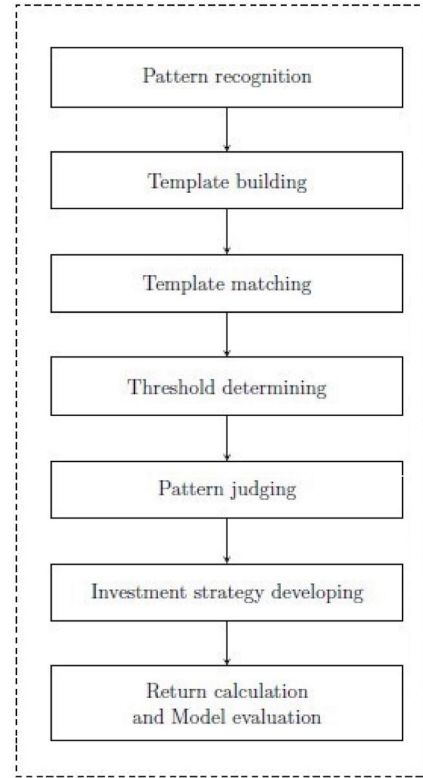


Fig. 7. Flowchart of proposed model.

Table 1. Pseudo code of PIP HS

Procedure head-and-shoulder pattern identification

Input: sequence $P[1 \dots m]$, fitting_window, holding_period

Output: candidate selected pattern CSP

Begin

For each data point $p[i]$ in P , between $p[i]$ and $p[i + \text{fitting_window} - 1]$

Set test point $TP[1] = p[i]$, $TP[7] = p[i + \text{fitting_window} - 1]$

Set the point $TP[3] = p[j]$, with maximum distance between $p[i]$ and $p[i + \text{fitting_window} - 1]$

Repeat until $TP[1..7]$ are all filled

Condition 1: $TP[4] > TP[2]$ and $TP[4] > TP[6]$

Condition 2: $TP[2] > TP[1]$ and $TP[2] > TP[3]$

Condition 3: $TP[6] > TP[5]$ and $TP[6] > TP[7]$

Condition 4: $TP[3] > TP[1]$

Condition 5: $TP[5] > TP[7]$

Condition 6: Closing price ($p[i + \text{fitting_window} - 1 + \text{holding_period}]$) > Closing price ($p[i + \text{fitting_window} - 1]$)

If all the conditions are met

Add $p[i]$ to CSP

End if

End for

Return CSP

Table 2. Pseudo code of PIP HS combine

Procedure DIF pattern identification

Input: sequence CSP[1...m], fitting_window, holding_period, distance_threshold

Output: Selected pattern SP

Begin

For each data point p[i] in CSP, between p[i+ fitting_window/2] and p[i+ fitting_window-1]

Set test point TP[1] = p[i+ fitting_window/2], TP[4] = p[i+ fitting_window -1]

Set test point TP[3] = p[j], with maximum distance between p[i+ fitting_window/2] and p[i+ fitting_window -1]

Repeat above process once again until TP[1,2,3,4] are all filled

Condition: TP[4]>TP[2]>TP[3]>TP[1]

If the condition is met

Add p[i] to SP

End if

End for

Return SP

Table 3. Pseudo code of template building

Pseudo code of template building

1) There are T-days daily closing stock prices ($P_1, P_2, \dots, P_T$) in a selected pattern, rank them in descending order, where T equals to the fitting-window size

2) Calculate $I_{t,i}$ for trading day t, ($t = 1, 2, \dots, T$ and $i = 1, 2, \dots, 10$)

$I_{t,i} = 1$, if $(i - 1) \cdot T/10 \text{Rank}(P_t) i \cdot T/10$

$I_{t,i} = 10$, otherwise

where, t represents the abscissa and i represents the ordinate of the template

3) Repeat the procedure 2) until all of the $I_{t,i}$ of n selected pattern is calculated

4) Calculate weights for every grid of the template for n selected combined patterns, where n equals to the number of possible combined patterns we recognized in 3.2.1

$$cp_w_{t,i} = \sum_{m=1}^n \text{Head_and_shoulderpattern}_m(I_{t,i}) \cdot \frac{RR_n}{TRR}$$

$$RR = (P_{t+\text{holding_period}} - P_t)/P_t$$

where RR represents return rate

$$= (\text{closingindex}_{t+\text{holding_period}} - \text{closingindex}_t)/\text{closingindex}_t$$

$$TRR \text{ represents the sum of return rate} = \sum_{m=1}^n RR_m$$

Table 4. Pseudo code of template matching

Pseudo code of template matching

1) There are T-days daily closing stock prices ($P_1, P_2, \dots, P_T$) in a selected pattern, rank them in descending order where T equals to the fitting-window size

2) Calculate $J_{t,i}$ for trading day t, ($t = 1, 2, \dots, T$ and $i = 1, 2, \dots, 10$)

$J_{t,i} = 1$ if $(i - 1) \cdot T/10 \text{Rank}(P_t) i \cdot T/10$

$J_{t,i} = 0$, otherwise

where, t represents the abscissa and i represents the ordinate of the template

3) Calculate a fitting value (CP_Fit) for a stock pattern based on the template of charting pattern.

$$CP_Fit = \sum_{t=1}^T \sum_{i=1}^{10} cp_w_{t,i} \cdot J_{t,i}$$

pseudocode of the perceptually important point identification method.

According to the pseudocode, we can intuitively understand that recognizing the head-and-shoulder bottom pattern through the PIP method is extracting seven PIPs in a period time series. These seven points must be consistent with the characteristics

of the head-and-shoulder bottom pattern. There should be one head and two shoulders and the head should be lower than the shoulder. We determine the candidate pattern from the historical data if these characteristics are met. After recognizing the head-and-shoulder pattern, the 5MA-20MA DIF pattern observation is necessary. Table 2 explicates the pseudocode.

3.1.2 Template building

Through the above PIP method, we gained candidate head-and-shoulder patterns from the historical data. In this step, our team used the weighted average method to build templates and represent these patterns. Table 3 lists the pseudocodes for the chart template.

3.1.3 Template matching

In this step, every possible combined pattern selected in 3.2.1 will get its fitting value, representing the degree of matching the weighted average pattern built in Section 3.2.2. These fit-

ting values were used to determine the threshold that determines whether a pattern is a combined pattern. The pseudo-code illustrating the calculation of the fitting value is as follows: Table 4.

3.1.4 Threshold determining

In Section 3.2.3, n fitting values for n candidate combined patterns selected were obtained. In this step, we calculate three thresholds for each template that judge whether a pattern is combined using these fitting values. The thresholds are shown in the below equations. As the criteria for judging

Table 5. Forecasting accuracy of Crude Oil, NASDAQ100 and S&P500

(a) Crude Oil						
Period	Holding period = 10		Holding period = 15		Holding period = 20	
	Accuracy (%)	N (trades)	Accuracy (%)	N (trades)	Accuracy (%)	N (trades)
2017	49.14	291	65.71	70	52.17	115
2018	61.11	19	62.96	27	84.62	13
2019	27.59	29	34.25	73	50	58
2020	58.46	65	74.29	35	75	28
Average Accuracy	49.63		55.61		56.54	
(b) NASDAQ100						
Period	Holding period = 10		Holding period = 15		Holding period = 20	
	Accuracy (%)	N (trades)	Accuracy (%)	N (trades)	Accuracy (%)	N (trades)
2017	56.25	64	40.91	66	42.06	107
2018	53.57	28	64	25	27.27	11
2019	55	40	54.35	46	52.27	44
2020	55.56	18	63.16	19	63.64	11
Average Accuracy	55.33		51.28		45.09	
(c) S&P500						
Period	Holding period = 10		Holding period = 15		Holding period = 20	
	Accuracy (%)	N (trades)	Accuracy (%)	N (trades)	Accuracy (%)	N (trades)
2017	46.99	83	42.86	28	64.71	34
2018	54.76	168	37.5	96	50.75	67
2019	0	7	0	5	33.33	9
2020	50.63	79	51.67	60	55.07	69
Average Accuracy	50.74		41.8		54.19	

stock patterns, the strictness of the three thresholds decreases in turn.

$$CP_Threshold_{max} = \max(C_Fit_m) \quad (6)$$

$$CP_Threshold_{avg} = \sum_{m=1}^n (CP_F_Fit_m) / n \quad (7)$$

$$CP_Threshold_{min} = \min(CP_Fit_m) \quad (8)$$

3.2 Investment strategy developing

After building a weighted average template and calculating the thresholds using training data, an investment strategy can be developed using the following procedure:

Step1: Select a period including T-day daily closing stock prices as a testing stock pattern, where T equals the fitting-window size.

Step2: Calculate the fit value for this testing period using the weighted average template obtained from the training data.

Step3: Compare the fit value with the threshold value. If the fit value is not less than the threshold value, the testing stock pattern can be recognized as a combined pattern. The next day is invited and the holding period is held.

4. Model Verification

We selected Crude Oil, S&P 500 and NASDAQ 100 as experimental datasets to execute evaluations for verifying the proposed model. The periods of these three indexes were all four years, from 2017/1/1 to 2020/12/31. Each index was divided into two parts, training, and testing, at a ratio of 4:1. In the practical experiment, the fitting window was set as 50 and the average value (Eq. 7) was used as the fitting value threshold. Three periods, 10, 15 and 20 are the factor holding period to observe the forecasting results.

The criterion for evaluating the performance of the proposed investment model is the forecasting accuracy rate, defined as follows:

$$\text{Forecasting accuracy rate} = \frac{N_{\text{positive_return}}}{N} \quad (9)$$

where $N_{\text{positive_return}}$ is the number of trading days with positive return after the holding period of the selected trading days, and N is the number of trading days selected by the model. Table 5 presents forecasting accuracy results.

5. Conclusions

This study proposes an investment model based on pattern recognition with moving average technical indicators for future markets. According to the forecasting results in Section 4, the forecasting accuracy rates of all three futures markets are more than 50% in a certain holding period. In some years, forecasting accuracy rates have reached 80%. Therefore, we acknowledge that this model has the guaranteed investment accuracy. As the investment model uses only a type of technical indicator moving average, the calculated amount is relatively small, and the time consumed for investment decision-making is short, which can better cope with the rapidly changing futures market. In addition, even if common investors do not have statistical or financial knowledge reserves, they can simply operate according to the model's instructions to ensure the investment's accuracy. The decision-making process and judgment criteria are intuitive and easy to understand. However, using a single type of technical indicator can reduce the calculated amount and the difficulty of understanding the methods. Still, the prediction accuracy and reliability of the model will also be reduced. Therefore, this method cannot match all head-and-shoulder patterns and determine the best holding period. In this model, we used a fixed-fitting window and a holding period. Here, patterns will be of different lengths of time, and a fixed-fitting window will not be able to find all the matching patterns. The fixed holding period will also make the model too rigid and unable to trade at a suitable time. In future research, the priority is to solve the problem of the fitting window to obtain a model with more potential investment opportunities. Subsequently, adding other types of technical indicators can be considered to obtain higher accuracy and profits.

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